

# Imprecise Probabilistic Machine Learning

**Winter Semester 2026/27**

Saarland University / University of Hamburg / Public Enrolment

**Lecture 3**: Gambles, Sets of Desirable Gambles, and Lower Prevision

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# Recap

1. Uncertainty in machine learning
2. Precise probability and its pitfalls
3. Beyond precise probability

# Outline

1. Gambles
2. Set of desirable gambles
3. Coherent lower prevision and lower probability



# From Outcomes to Random Variables



- Consider rolling a fair six-sided die: $\Omega = \{1, 2, 3, 4, 5, 6\}$.
- Define a **random variable** $X$: $\Omega \to \mathbb{R}$, $X(\omega) = \omega$.

| outcome $\omega$ | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|
| $X(\omega)$ | 1 | 2 | 3 | 4 | 5 | 6 |

- The random variable then induces a **probability distribution**:

$$P_X(B) = \mathbb{P}(X \in B) = \mathbb{P}(X^{-1}(B))$$

- For a fair die,

$$P_X(X = k) = \frac{1}{6}, \qquad k = 1, \dots, 6$$

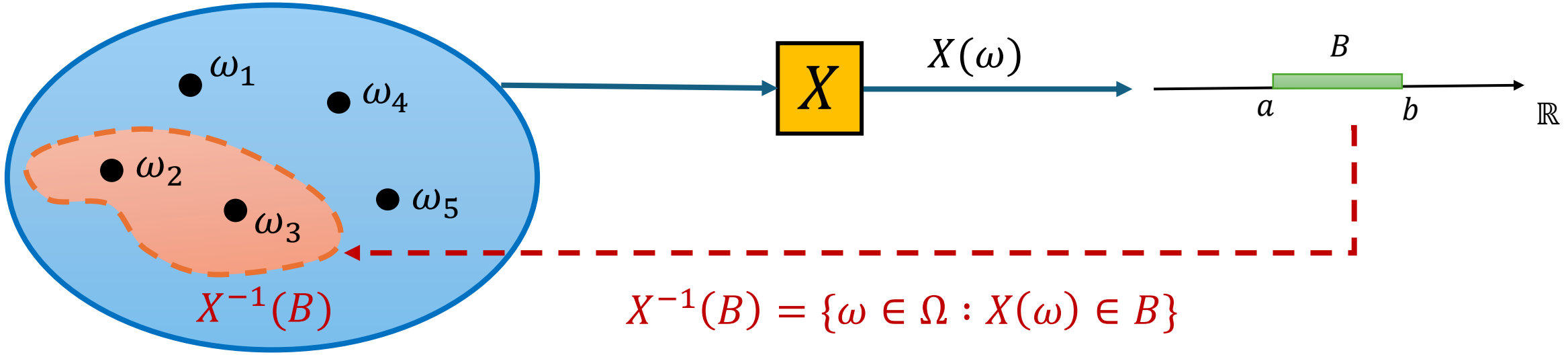
- A random variable is a function whose probability distribution $P_X$ is induced by the underlying probability measure $\mathbb{P}$.

# Random Variables

**Probability Space**
$(\Omega, \mathcal{F}, \mathbb{P})$

**Random Variable**
$\mathrm{X}\colon\ \Omega \to \mathbb{R}$

**Induced Distribution**
$P_X(B) = \mathbb{P}(X \in B) = \mathbb{P}(X^{-1}(B))$




$$X^{-1}(B) = \{\omega \in \Omega : X(\omega) \in B\}$$

# Indicator Gamble

- **Bet**: You receive €1 if the die shows an even number, and €0 otherwise.
- Let $A = \{2, 4, 6\}$. The corresponding **indicator gamble** is

$$1_A(\omega) = \begin{cases} 1, & \omega \in A \\ 0, & \omega \notin A \end{cases}$$

| Outcome $\omega$ | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|
| $1_A(\omega)$ | 0 | 1 | 0 | 1 | 0 | 1 |

- The expected payoff is

$$\mathbb{E}_P[1_A] = P(A) = \frac{1}{2}$$

- An indicator gamble represents the payoff of a **binary event**.

# General Gamble

- **Bet**: You receive €5 if the die shows 6, lose €2 if it shows 1, and and receive €0 otherwise.

$$f(\omega) = \begin{cases} — & \omega = 1 \\ — & \omega = 6 \\ — & \text{otherwise} \end{cases}$$

| Outcome ω | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|
| $f(\omega)$ | -2 | 0 | 0 | 0 | 0 | 5 |

- The expected payoff under the fair-die model is $\mathbb{E}_P[f] = \frac{-2+5}{6} = 0.5$.
- In IP, a gamble is defined as a **bounded real-valued function** on $\Omega$.

# Behaviour Foundations of IP

| Concept | Mathematical Object | Interpretation |
|---|---|---|
| Random variable $X$ | $X: \Omega \to \mathbb{R}$ | Quantity of interest |
| Indicator gamble | $1_A: \Omega \to \{0,1\}$ | Payoff associated with an event |
| **Gamble $f$** | $\boldsymbol{f: \Omega \to \mathbb{R}}$ | **State-dependent reward or loss** |
| Probability measure $P$ | $P: \mathcal{E} \to [0,1]$ | Beliefs about events |
| Distribution $P_X$ | $P_X(B) = P(X \in B)$ | Probabilities of value of $X$ |

In IP, ***gambles are treated as primitive object*** representing uncertain rewards or losses, without requiring a probability measure.

# General Gamble

| | Random Variable | Indicator Gamble | General Gamble |
|---|---|---|---|
| **Function** | $X: \Omega \to \mathbb{R}$ | $1_A: \Omega \to \{0,1\}$ | $f: \Omega \to \mathbb{R}$ |
| **Interpretation** | Numerical outcome | Whether an event occurs | Payoff or reward |
| **Example** | Face value | Die shows an even number | Reward for each outcome |
| **Values** | 1,2,3,4,5,6 | 0,1,0,1,0,1 | -3,2,-1,4,0,8 |

Events $P(A)$ → Indicator gambles $\mathbb{E}_P[1_A]$ → General gambles $\mathbb{E}_P[f]$

Expectations of gambles provide ***a richer language for reasoning under uncertainty***

# Gambles in Machine Learning

| Machine learning concept | Interpretation as a gamble | |
|---|---|---|
| Loss function | $f_h(z) = \ell(h, z)$ | $h = \text{model},\ z = \text{data}$ |
| Classification error | $f_h(x, y) = 1_{\{h(x) \neq y\}}$ | $x = \text{input},\ y = \text{output}$ |
| Reward function | $f_\pi(\omega) = R(\pi, \omega)$ | $\pi = \text{policy}$ |
| Utility function | $f_a(\omega) = u(a, \omega)$ | $a = \text{action}$ |
| Prediction error | $f_h(x, y) = (h(x) - y)^2$ | |
| Treatment benefit | $f(\omega) = Y_1(1, \omega) - Y_0(0, \omega)$ | $Y_t = \text{potential outcomes}$ |

**Exercise**: Think of any other concepts in machine learning that can be interpreted as a gamble.

# Precise Probabilities in Machine Learning

1. **Statistical Learning** – Minimizing Expected Risk
   Assume an underlying data-generating distribution $\boldsymbol{P}$.

$$h^* = \operatorname{argmin}_{h \in \mathcal{H}} \mathbb{E}_{Z \sim \boldsymbol{P}}[\ell(h, Z)], \qquad \hat{h} = \operatorname{argmin}_{h \in \mathcal{H}} \frac{1}{n} \sum_{i=1}^{n} \ell(h, z_i)$$

2. **Bayesian Learning** – Updating Beliefs
   Start with a prior probability distribution over parameters $\boldsymbol{P}(\theta)$.

$$P(\theta \mid D) = \frac{P(D \mid \theta)\boldsymbol{P}(\theta)}{P(D)}, \qquad P(y|x, D) = \int P(y|x, \theta) P(\theta|D)\, d\theta$$

# What if probabilities are *unknown*?

Suppose the die may be biased, but probabilities are unknown.

| Outcome ω | ⚀ | ⚁ | ⚂ | ⚃ | ⚄ | ⚅ |
|---|---|---|---|---|---|---|
| Probability | ? | ? | ? | ? | ? | ? |
| $f(\omega)$ | -2 | -1 | 0 | 2 | 3 | 5 |
| $g(\omega)$ | -3 | -2 | -1 | 1 | 2 | 4 |

**1) Expected utility**

$$\mathbb{E}_P[f] \text{ and } \mathbb{E}_P[g]$$

Require probability distribution

**2) Payoff bounds**

$$-2 \leq f(\omega) \leq 5$$
$$-3 \leq g(\omega) \leq 4$$

Bounded payoffs

**3) Dominance**

$$f(\omega) \geq g(\omega)$$

Better in every possible state

**Takeaway**: We can reason about gambles without knowing $P$.

# Acceptability

- Suppose someone offers you this gamble. Would you accept it?

$$f(\omega) = \begin{cases} 5, & \omega = 6 \\ -1, & \text{otherwise} \end{cases}$$

- How about these gambles?

$$f = (1,0,1,0,1,0), g = (-1,-1,-1,-1,-1,-1), h = (3,-8,0,1,-2,10)$$

- If you accept $f = (1,2,3,-1,-2,-3)$, should you also accept this gamble?

$$g = (2,4,6,-2,-4,-6)$$

- Instead of requiring you to specify a probability distribution, we can ask which gambles $f \in \mathbb{D} \subseteq \mathcal{L}(\Omega)$ you consider acceptable.

# Coherent Set of Acceptable Gambles

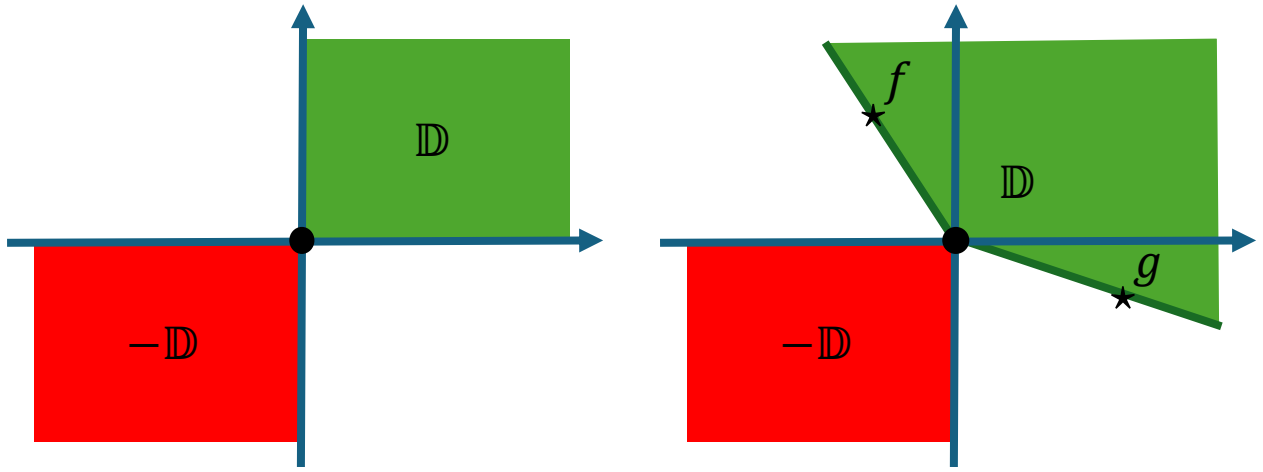
A set of **acceptable gambles** $\mathbb{D} \subseteq \mathcal{L}(\Omega)$ is *coherent* if and only if:

**(A1)** If $f < 0$, then $f \notin \mathbb{D}$ (avoid partial loss)

**(A2)** If $f \geq 0$, then $f \in \mathbb{D}$ (accept partial gain)

**(A3)** If $f \in \mathbb{D}$, then $\lambda f \in \mathbb{D}$ $(\forall \lambda \in \mathbb{R}, \lambda > 0)$ (positive scale invariance)

**(A4)** If $f \in \mathbb{D}$ and $g \in \mathbb{D}$, then $f + g \in \mathbb{D}$ (combination)

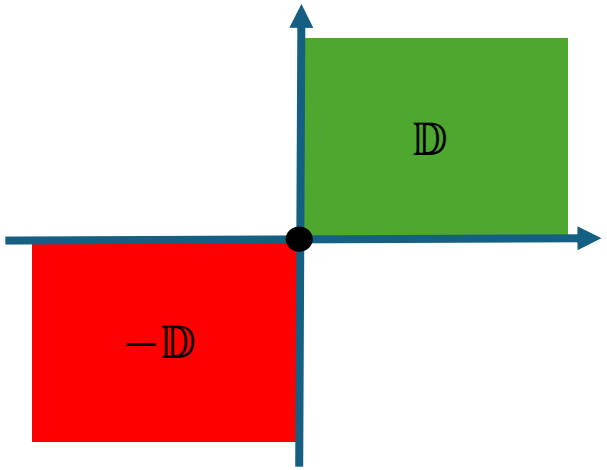



**Exercise:** Show that these four axioms entail a *monotonicity* property:

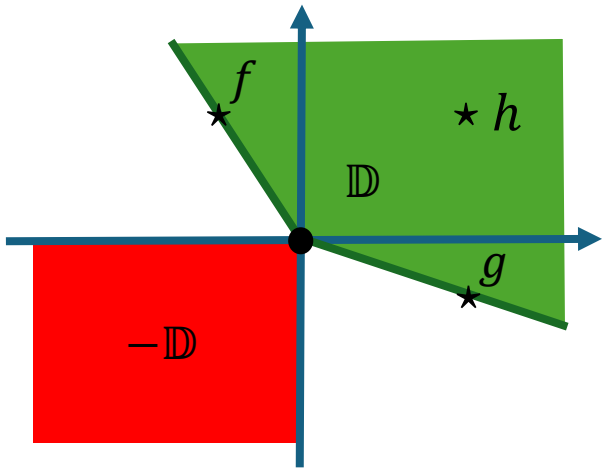
**(A5)** If $f \in \mathbb{D}$ and $g \geq f$, then $g \in \mathbb{D}$

That is, any bounded gamble $g$ that dominates an acceptable gamble $f$ is an acceptable gamble.

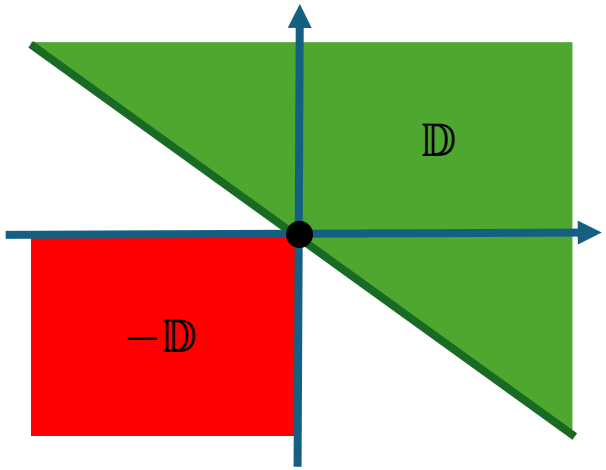
# Coherent Set of Acceptable Gambles




Minimal commitment




$\mathbb{A} = \{f, g, h\}$




Maximal commitment

# Lower Prevision

Let $\mathbb{D} \subseteq \mathcal{L}(\Omega)$ be a set of acceptable gambles. Then, for any gamble $f \in \mathcal{L}(\Omega)$ and real number $\mu$, the **lower prevision** operator $\underline{P}(\cdot)$ is

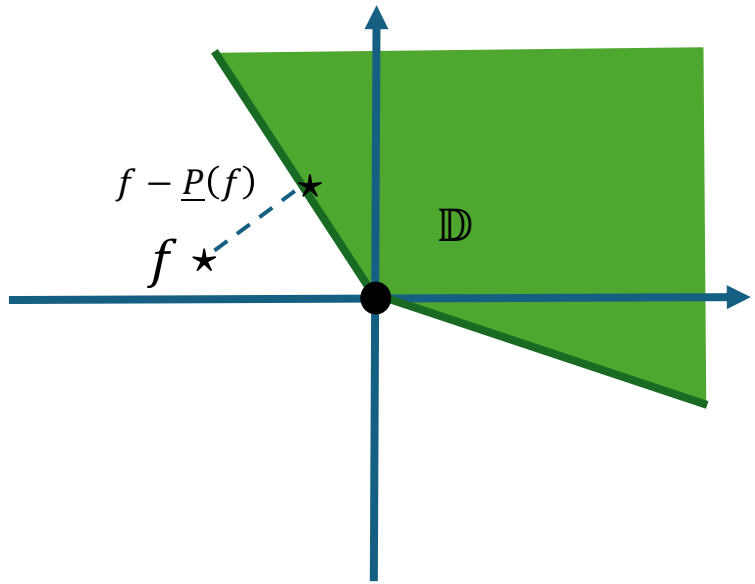
$$\underline{P}(f) := \sup\{\mu \in \mathbb{R} : f - \mu \in \mathbb{D}\}$$

**Interpretations**:

- The *supremum price* $\mu$ you are willing to pay for $f$.
- For all $\varepsilon > 0$, you are willing to accept the *uncertain reward*

$$f - \underline{P}(f) + \varepsilon$$

where $\underline{P}(f)$ and $\varepsilon$ are constant gambles.



# Upper Prevision

Let $\mathbb{D} \subseteq \mathcal{L}(\Omega)$ be a set of acceptable gambles. Then, for any gamble $f \in \mathcal{L}(\Omega)$ and real number $\lambda$, the **upper prevision** operator $\overline{P}(\cdot)$ is

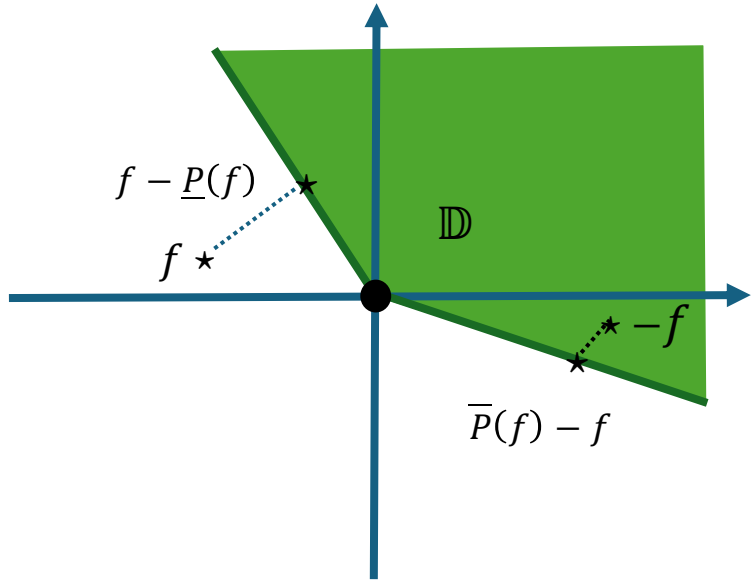
$$\overline{P}(f) := \inf\{\lambda \in \mathbb{R} : \lambda - f \in \mathbb{D}\}$$

**Interpretations**:

- The *infimum selling price* $\lambda$ for $f$ that is acceptable to you.
- For all $\varepsilon > 0$, you are willing to accept the *uncertain reward*

$$\overline{P}(f) - f + \varepsilon$$

where $\overline{P}(f)$ and $\varepsilon$ are constant gambles.



# Lower and Upper Previsions

$$\boxed{\overline{P}(f) = -\underline{P}(-f)}$$

By the definition of upper prevision,

$$\begin{aligned}
\overline{P}(f) &:= \inf\{\, \lambda \in \mathbb{R} : \lambda - f \in \mathbb{D} \,\} && \text{(by definition)} \\
&= \inf\{\, \lambda \in \mathbb{R} : -f - (-\lambda) \in \mathbb{D} \} \\
&= \inf\{\, -\mu \in \mathbb{R} : -f - \mu \in \mathbb{D} \,\} && \text{(setting } \mu = -\lambda\text{)} \\
&= -\sup\{\, \mu \in \mathbb{R} : -f - \mu \in \mathbb{D} \,\} && \text{(}\inf\{-a\} = -\sup\{a\}\text{)} \\
&= -\underline{P}(-f) && \text{(by definition)}
\end{aligned}$$

Unlike precise probabilities, self-conjugacy does not generally hold.

# Coherent Lower Prevision

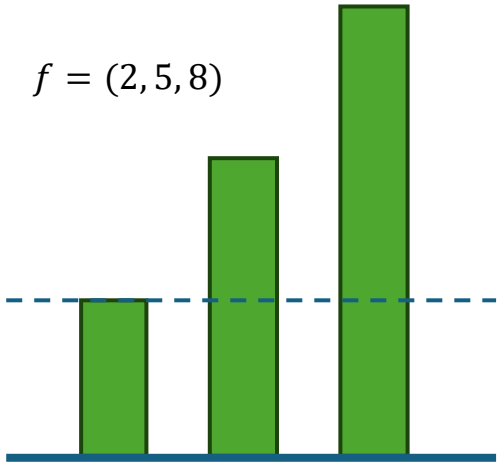
A **lower prevision model** $(\Omega, \mathbb{D}, \underline{P})$ where $\underline{P}$ is a real functional called a lower prevision. For gambles $f, g \in \mathcal{L}(\Omega)$ and any strictly positive number $\lambda$, $\underline{P}$ is a *coherent lower prevision* if and only if the following conditions are satisfied:

**(LP1).** $\underline{P}(f) \geq \inf_{\omega \in \Omega} f(\omega)$ (bounds)
**(LP2).** $\underline{P}(f + g) \geq \underline{P}(f) + \underline{P}(g)$ (super-additivity)
**(LP3).** $\underline{P}(\lambda f) = \lambda \underline{P}(f)$ (non-negative homogeneity)

| Property | Key Intuition |
|---|---|
| **LP1**: Bounds | Respect guaranteed rewards |
| **LP2**: Super-additivity | Combining gambles preserves acceptable prices |
| **LP3**: Non-negative homogeneity | Value scales with the stakes |

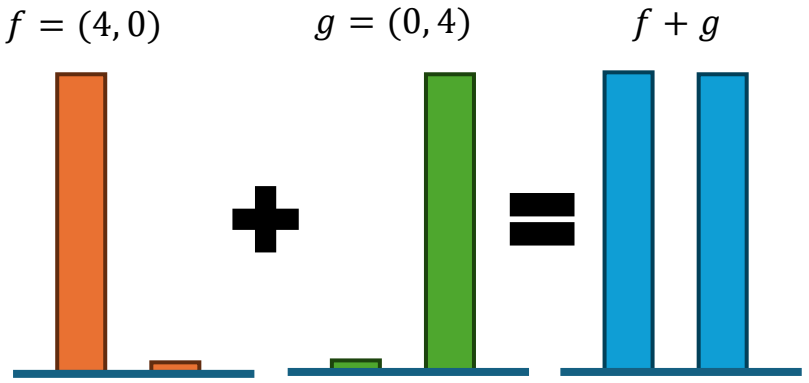
# Coherent Lower Prevision

**LP1: Bounds**
$\underline{P}(f) \geq \inf_{\omega \in \Omega} f(\omega)$



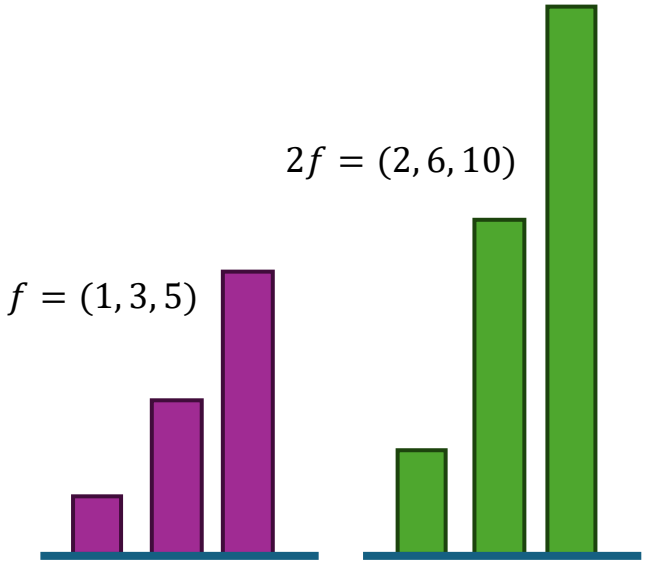

If you are guaranteed to receive at least €2, you should be willing to pay any price below €2.

**LP2: Super-additivity**
$\underline{P}(f + g) \geq \underline{P}(f) + \underline{P}(g)$




Two individually uncertain gambles can complement each other, producing a more valuable combined gamble.

**LP3: Non-negative homogeneity**
$\underline{P}(\lambda f) = \lambda \underline{P}(f)$




Doubling the stakes doubles the acceptable buying price.

# Coherent Lower Prevision

A lower prevision $\underline{P}$ should ***avoid sure loss***, meaning that

$$\sup_{\omega\in\Omega} \sum_{i=1}^{n} [f_i(\omega) - \underline{P}(f_i)\,] \geq 0 \quad \text{for all } n \geq 0 \text{ and all } f_1, \dots, f_n$$

If this is not satisfied, then there are $n > 0, f_1, \dots, f_n$ and $\delta > 0$ such that

$$\sup_{\omega\in\Omega} \sum_{i=1}^{n} [f_i(\omega) - (\underline{P}(f_i) - \delta)] \leq -\delta$$

**Conclusion**: the sum of desirable transactions $f_i(\omega) - (\underline{P}(f_i) - \delta)$ results in a loss of at least $\delta$, no matter the outcome of the experiment.

# **Coherent** Lower Prevision

A stronger rationality condition, ***coherence***, requires that supremum acceptable buying price for a gamble $f$ cannot be raised by considering a positive linear combination of a finite number of other acceptable gambles:

$$\sup_{\omega\in\Omega} \sum_{i=1}^{n} \left[f_i(\omega) - \underline{P}(f_i)\right] - m\left[f_0(\omega) - \underline{P}(f_0)\right] \geq 0$$

for all natural $n, m \geq 0$ and all $f_0, f_1, \ldots, f_n$.

**Exercise**: What if the lower prevision is incoherent?

# Linear Prevision

**Exercises:**
1. If a gamble $f$ costs €2, what should twice the gamble cost? What about the opposite gamble $-f$?
2. Show that the *homogeneity* condition:
   **(P3)** $P(\lambda f) = \lambda P(f),\ \lambda \in \mathbb{R}$
   follows from P1 and P2.

If $P$ is a **linear prevision** on a linear space $\mathbb{D}$ of bounded gambles, then $P$ is *coherent* if and only if the following conditions are satisfied:

**(P1).** $P(f) \geq \inf_{\omega \in \Omega} f(\omega)$ (bounds)
**(P2).** $P(f+g) = P(f) + P(g)$ (additivity)

## Interpretation:

- $P(f)$ assigns a **fair price** to every gamble $f$.
- A coherent linear prevision is an *expectation* under a precise (finitely additive) probability measure.

$$P(f) = \mathbb{E}_{\mathbb{P}}[f]$$

| Portfolio | Linear Prevision $P$ | | |
|---|---|---|---|
| $f = (0,2)$ | €0.80 | €0.80 | €0.80 |
| $g = (2,0)$ | €1.20 | €1.20 | €1.20 |
| $f + g = (2,2)$ | **€2.00** | **€1.50** | **€2.50** |
| **Profit** | **€0.00** | **€0.50** | **€0.50** |

**No-arbitrage**: Coherent linear previsions assign prices that cannot be exploited through risk-free trades.

# Coherent Lower Previsions: Characterisation

- **Domination**: A linear prevision $P$ *dominates* a lower prevision $\underline{P}$ if

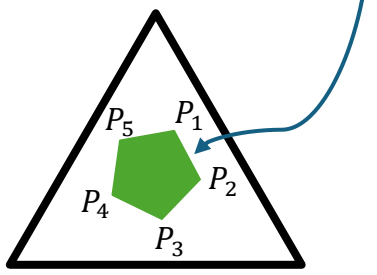
$$P(f) \geq \underline{P}(f), \qquad \forall f \in \mathbb{D}$$

*A dominating linear prevision assigns at least as high a value to every gamble.*

- **Lower-envelope representation**: A lower prevision is coherent iff it is the lower envelope of a closed, convex set of dominating linear previsions:

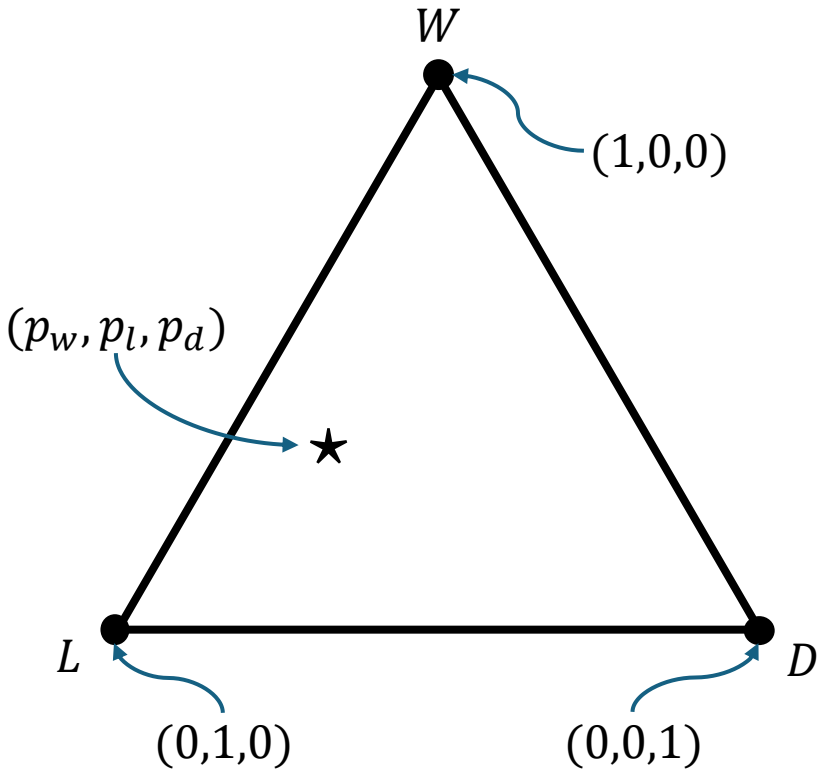
$$\underline{P}(f) = \inf_{P \in \mathcal{M}[\underline{P}]} P(f), \qquad \mathcal{M}[\underline{P}] = \{P \in \mathcal{P} : P(f) \geq \underline{P}(f), \forall f \in \mathbb{D}\}$$

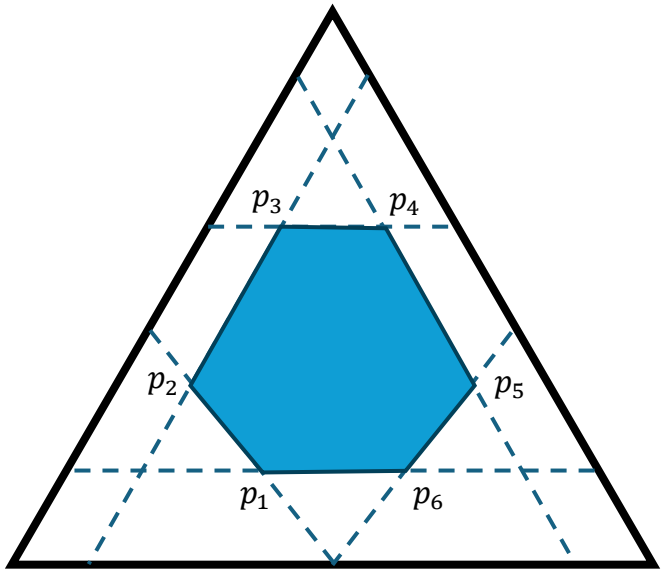
$\Omega = \{\omega_1, \omega_2, \omega_3\}$
$p_1 + p_2 + p_3 = 1, p_i \geq 0$

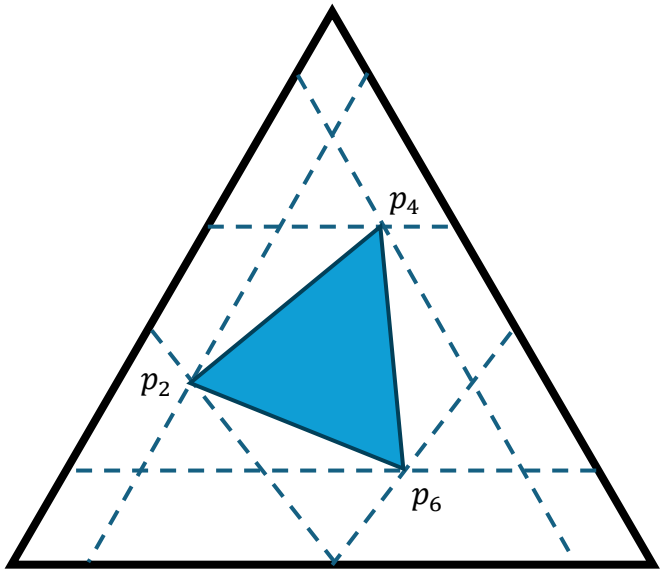


# Credal Set

Consider the possibility space $\Omega = \{\text{win, lose, draw}\}$.



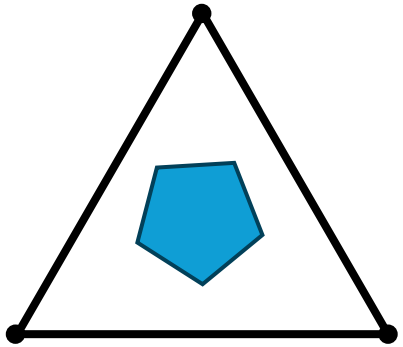



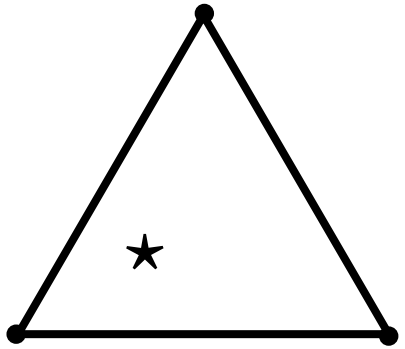
# Lower and Upper Previsions

- Let $\mathcal{C}$ be a non-empty subset of a set of all probabilities, i.e., $\mathcal{C} \subseteq \mathcal{P}$. The **lower and upper previsions** can be defined as

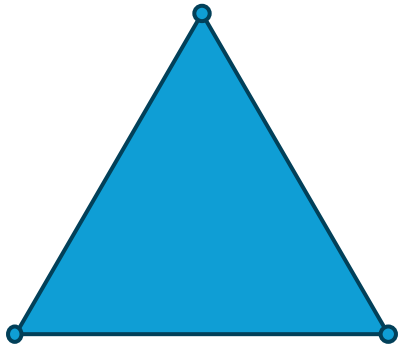
$$\underline{P}(f) \coloneqq \inf_{P \in \mathcal{C}} \mathbb{E}_P[f], \qquad \overline{P}(f) \coloneqq \sup_{P \in \mathcal{C}} \mathbb{E}_P[f]$$



$\mathcal{P}^A = \{ P \in \mathcal{P} : P(A) = 1 \}$



$\mathcal{C} = \{P\}$



$\mathcal{C} = \mathcal{P}$

# Probability Intervals

A **probability interval** $(\underline{p}, \overline{p})$ is a pair of lower and upper probability mass functions satisfying:

1. $0 \leq \underline{p}_\omega \leq \overline{p}_\omega \leq 1, \forall \omega \in \Omega$ (bounded)
2. $\sum_{\omega \in \Omega} \underline{p}_\omega \leq 1 \leq \sum_{\omega \in \Omega} \overline{p}_\omega$ (proper)
3. $\underline{p}_\omega \geq 1 - \sum_{z \neq \omega} \overline{p}_z$ and
   $\overline{p}_\omega \leq 1 - \sum_{z \neq \omega} \underline{p}_z$ for all $\omega \in \Omega$ (reachable)

A set of compatible pmfs is called the **credal set**:

$$\mathcal{C} = \{p : \underline{p}_x \leq p_x \leq \overline{p}_x\}$$

# Reachability

Assume probability intervals for three outcomes: $\Omega = \{\omega_1, \omega_2, \omega_3\}$

**Reachable**

$$p_1 \in [0.2, 0.5]$$
$$p_2 \in [0.3, 0.6]$$
$$p_3 \in [0.1, 0.4]$$

The intervals satisfy the boundedness and properness conditions.

But consider the lower bound $p_1 = 0.2$. Since probabilities must sum to one, the remaining outcomes must satisfy $p_2 + p_3 = 0.8$. This is possible because their upper bounds sum to $0.6 + 0.4 = 1$. Similarly, the upper bound $p_1 = 0.5$ is attainable because the remaining outcomes can sum to $0.5$.

**Unreachable**

$$p_1 \in [0.1, 0.5]$$
$$p_2 \in [0.2, 0.4]$$
$$p_3 \in [0.2, 0.4]$$

These intervals satisfy boundedness and properness:

$$\sum_i \underline{p}_i = 0.5 \leq 1, \qquad \sum_i \overline{p}_i = 1.3 \geq 1$$

However, the lower bound $p_1 = 0.1$ is **impossible** because $p_1 = 0.1 \Longrightarrow p_2 + p_3 = 0.9$, but $p_2 + p_3 \leq 0.4 + 0.4 = 0.8$. Therefore, $p_1 \geq 0.2$ for every compatible probability distribution.

# Lower and Upper Probabilities

The lower and upper probabilities can be defined as

$$\underline{P}(S) := \max\left\{\sum_{\omega\in S}\underline{p}_\omega\,, 1-\sum_{\omega\in S^c}\overline{p}_\omega\right\}, \qquad \overline{P}(S) := \min\left\{\sum_{\omega\in S}\overline{p}_\omega\,, 1-\sum_{\omega\in S^c}\underline{p}_\omega\right\}$$

**Exercise**: Consider the following lower and upper pmf:

| | $X$ | $Y$ | $XY$ | $YY$ |
|---|---|---|---|---|
| $\overline{p}$ | 3/8 | 3/8 | 5/8 | 5/8 |
| $\underline{p}$ | 1/8 | 1/8 | 3/8 | 0/8 |

Then, calculate:

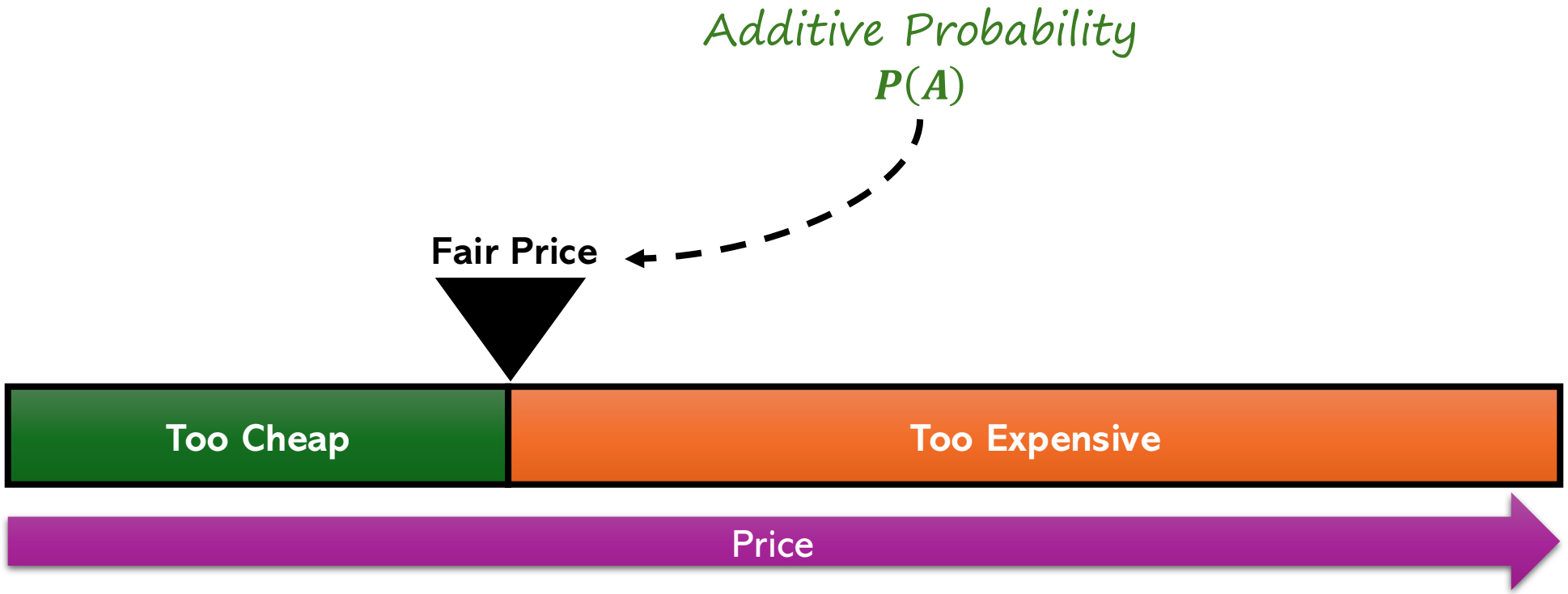
- $\overline{P}(\{Y, XY\}) = ?$
- $\underline{P}(\{X, Y\}) = ?$

# Probability Judgement

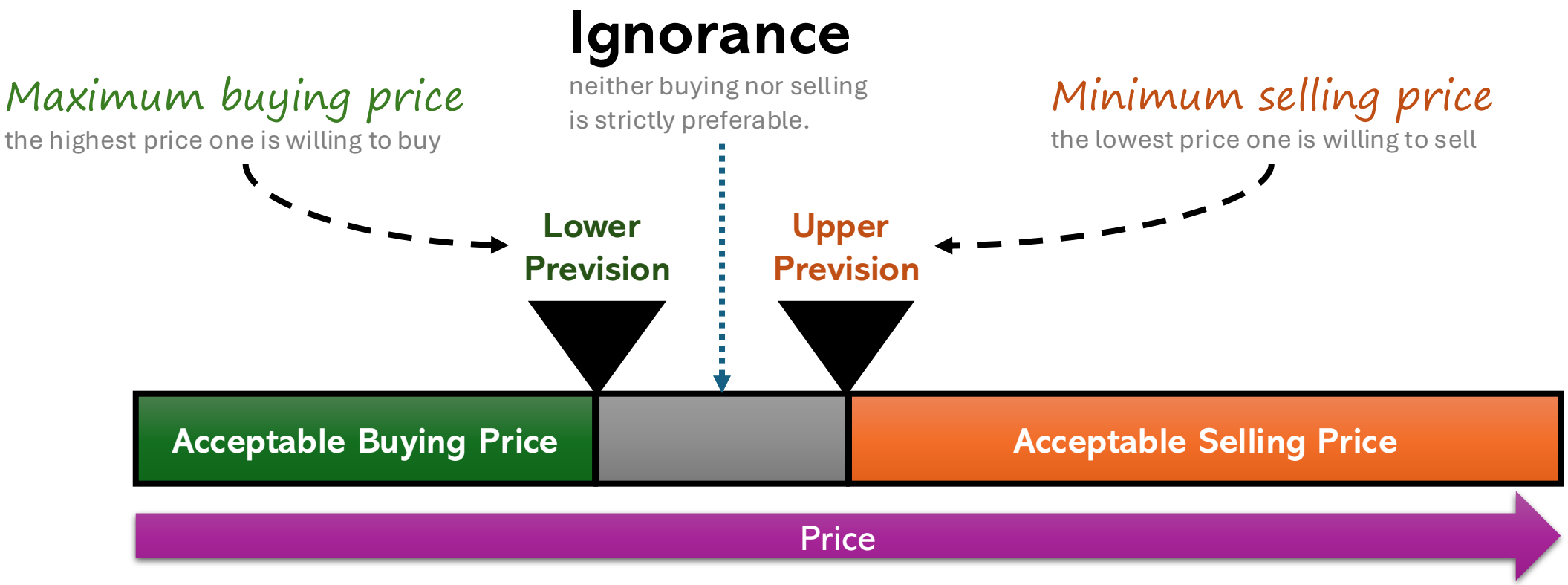
| Probability Judgement | Acceptable Gamble | Lower Prevision | |
|---|---|---|---|
| $E$ is probable | $E - \frac{1}{2} \in \mathbb{D}$ | $\underline{P}(E) \geq \frac{1}{2}$ | |
| $E$ is at least as probable as $F$ | $E - F \in \mathbb{D}$ | $\underline{P}(E - F) \geq 0$ | $*$ |
| $E$ is twice as probable as $F$ | $E - 2F \in \mathbb{D}$ | $\underline{P}(E - 2F) \geq 0$ | $*$ |
| Conditional on $G$, $E$ is at least as probable as $F$ | $G(E - F) \in \mathbb{D}$ | $\underline{P}(G(E - 2F)) \geq 0$ | $*$ |
| The odds against $E$ are at least 3 to 1 | $E^c - 3E \in \mathbb{D}$ | $\overline{P}(E) \leq \frac{1}{4}$ | |
| $E$ without $F$ is improbable | $\frac{1}{2} - E + F \in \mathbb{D}$ | $\overline{P}(E - F) \leq \frac{1}{2}$ | $*$ |
| Your degree of belief that $E$ is $\frac{1}{3}$ | $E - \frac{1}{3} \in \mathbb{D}$ and $\frac{1}{3} - E \in \mathbb{D}$ | $\underline{P}(E) = \frac{1}{3}$ and $\overline{P}(E) = \frac{1}{3}$ | |

The comparative probability judgments marked by $*$ cannot be modeled by lower probabilities.

# Big Picture


Additive Probability
$P(A)$
Fair Price
Too Cheap
Too Expensive
Price

# Big Picture


Ignorance
neither buying nor selling is strictly preferable.
Maximum buying price
the highest price one is willing to buy
Minimum selling price
the lowest price one is willing to sell
Lower Prevision
Upper Prevision
Acceptable Buying Price
Acceptable Selling Price
Price

# Big Picture

From events to arbitrary gambles

From precise to imprecise assessment

| | **Events/Indicator Gamble** | **General Gambles** |
|---|---|---|
| **Precise** | Probability $P(A)$ | Linear prevision $P(f)$ |
| **Imprecise** | Lower probability $\underline{P}(A)$ | Lower prevision $\underline{P}(f)$ |

Is *imprecision* relevant to modern machine learning?

*Distribution shift · Compositional generalization · Continual learning · Multi-agent systems · Causal inference · Open-ended environments · AI for scientific discovery · AI safety · …*

**Imprecision** is an inherent feature of learning under uncertainty.